

Cornell CS 3/5780

Introduction to Machine Learning

Supervised Learning

Instructors: Karthik Sridharan and John Thickstun



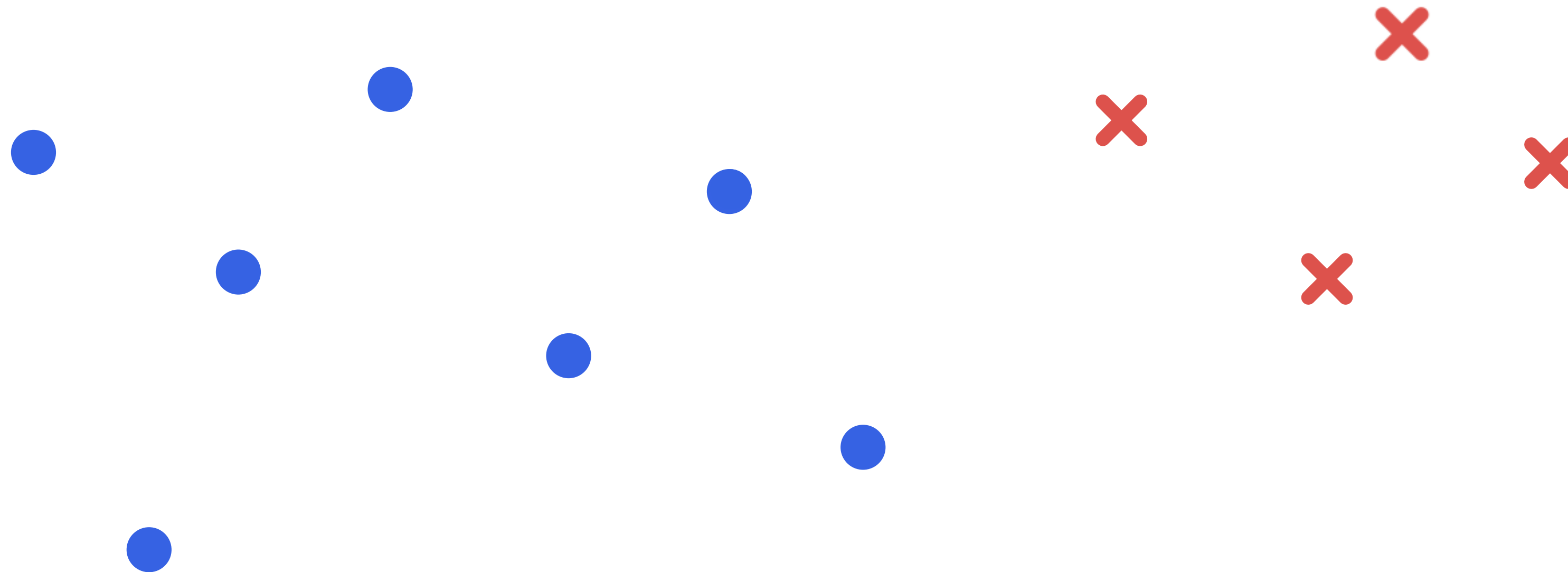
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College of Computing + Information Science



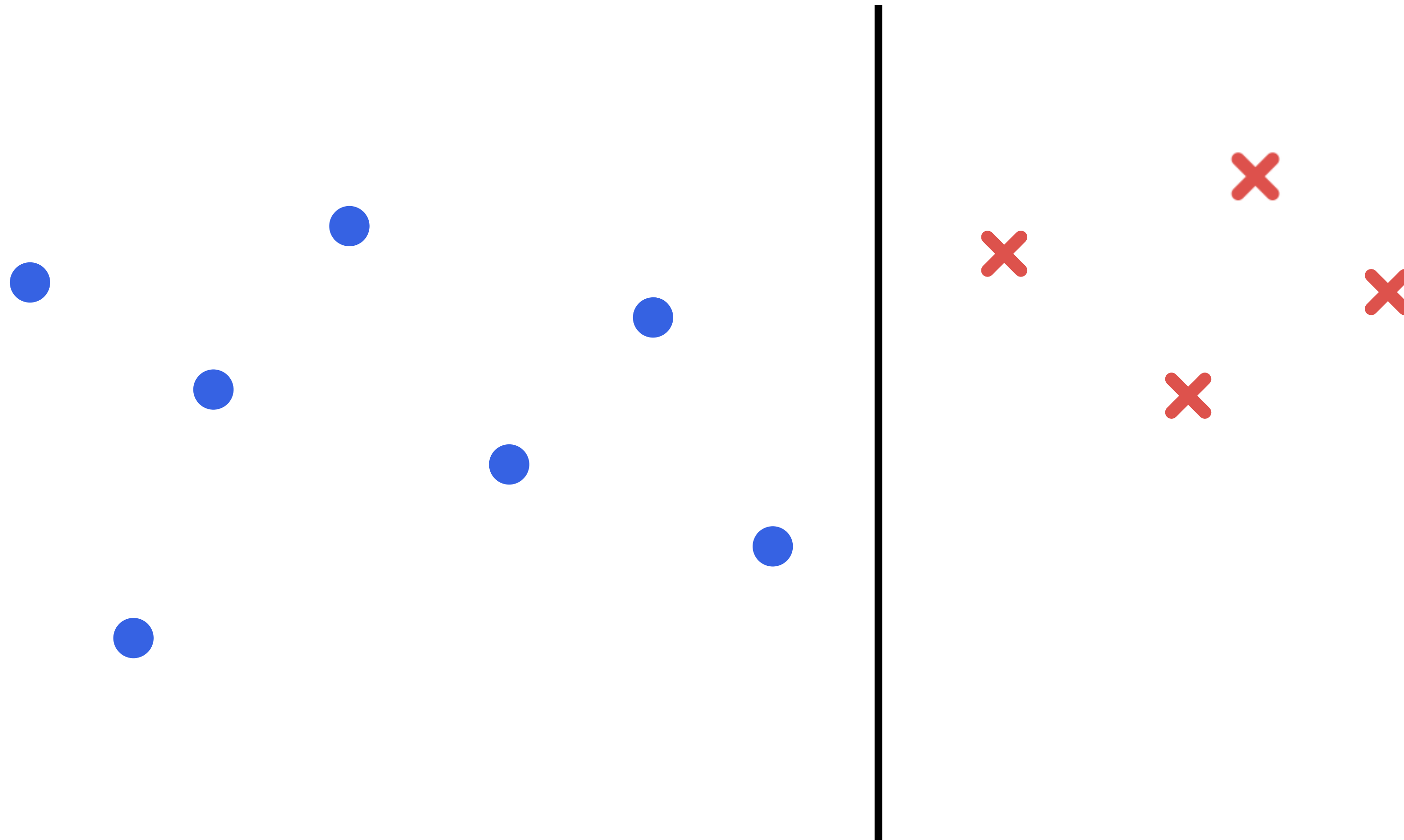
CIS Partner Finding Social

Thursday, September 3rd
4:30 – 5:30 PM
Gates 114

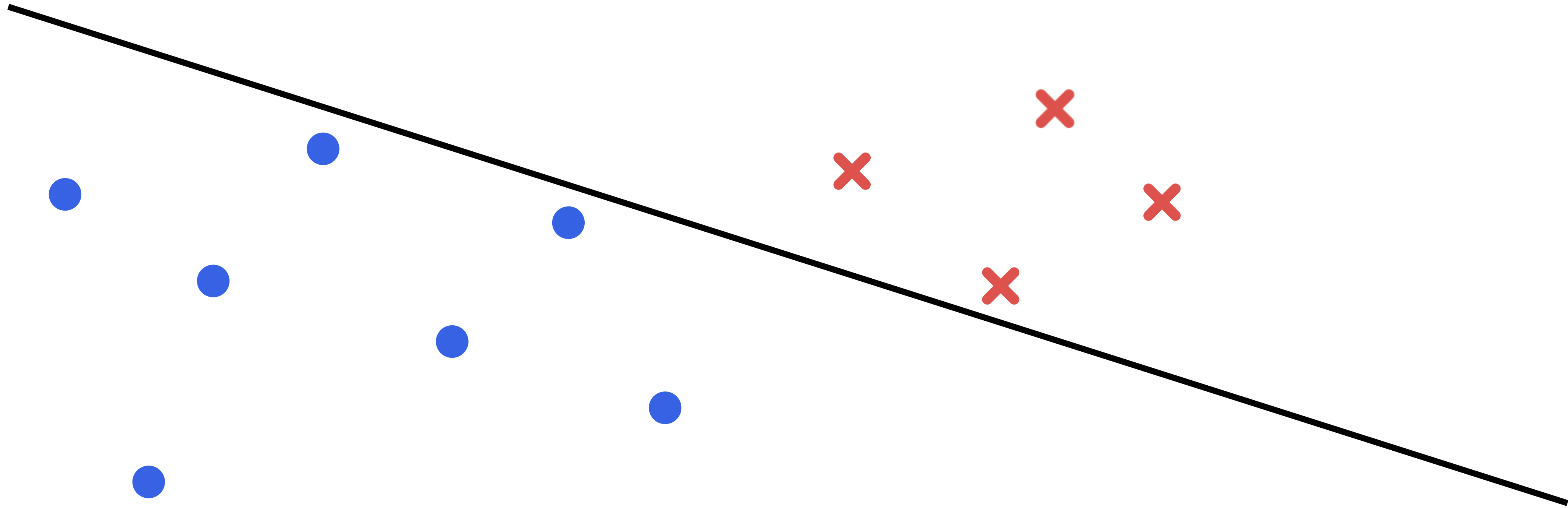
Linear Classifiers



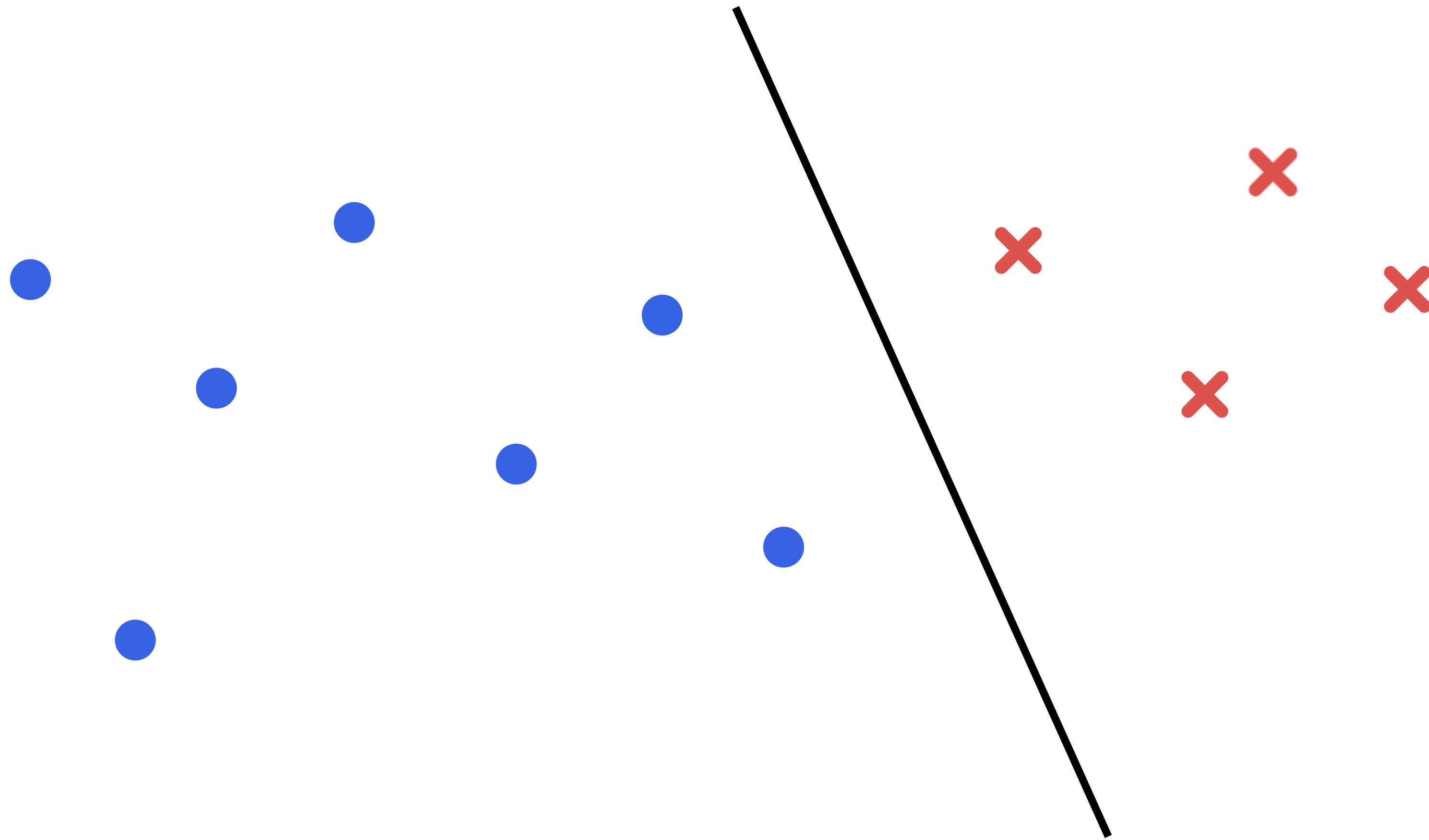
Linear Classifiers



Linear Classifiers

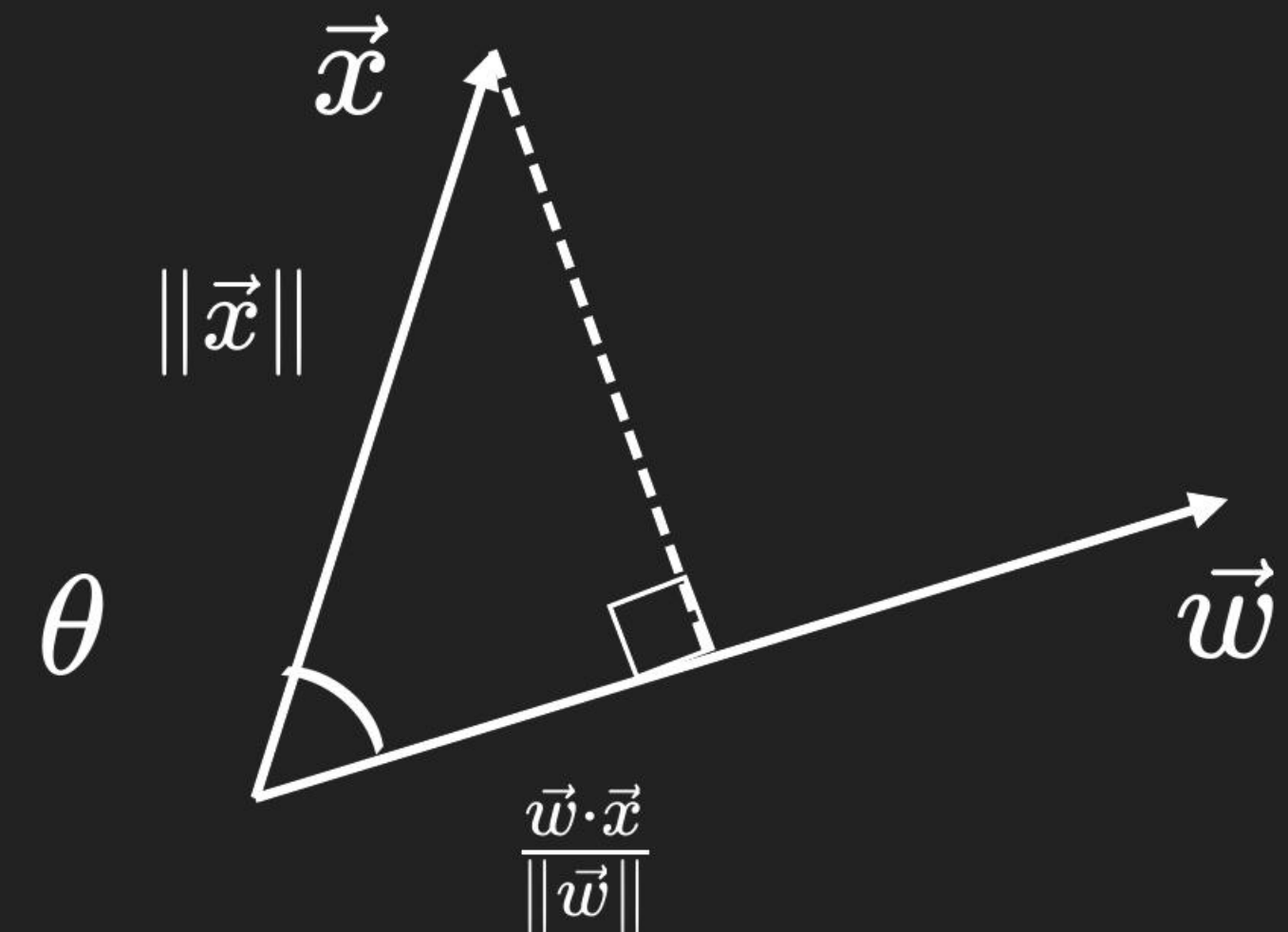


Linear Classifiers



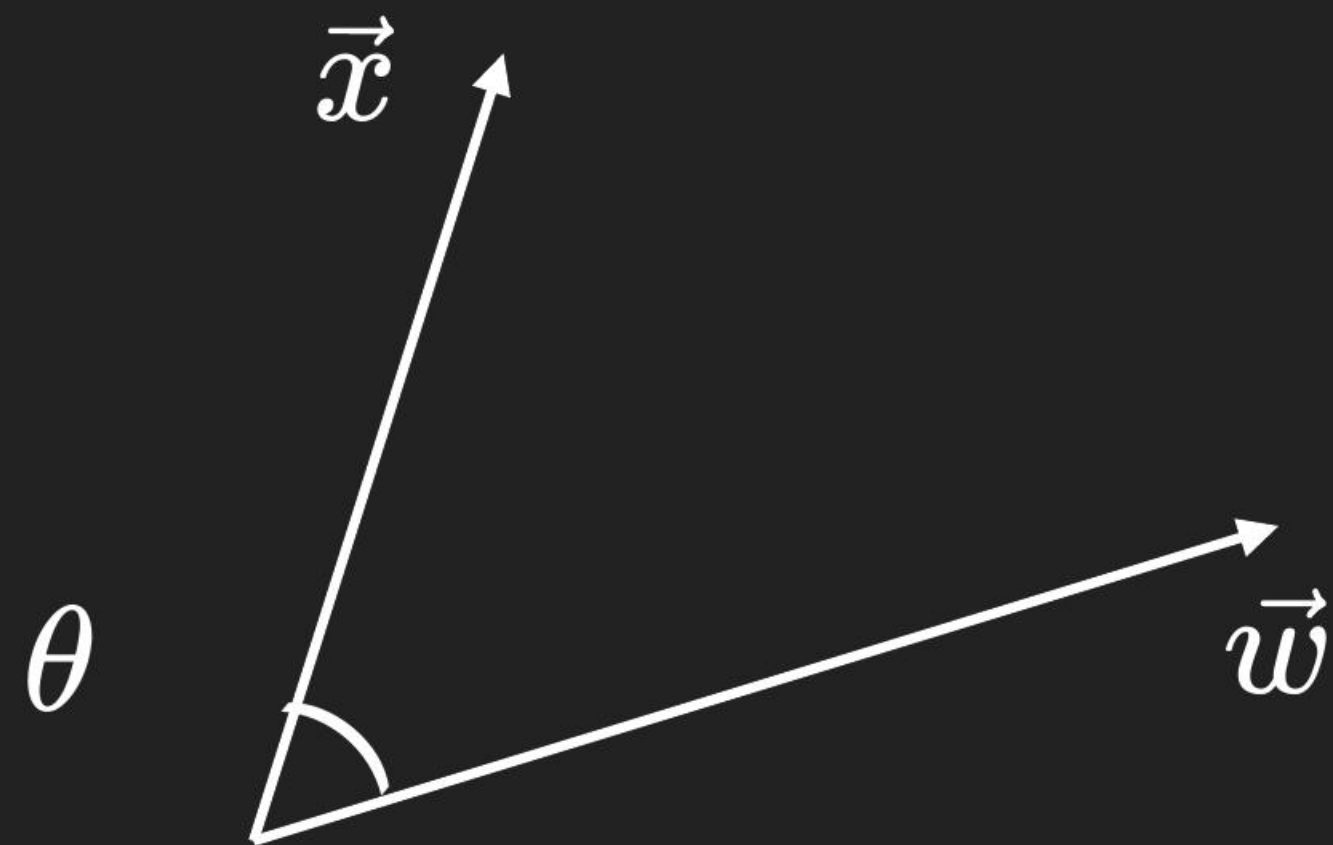
Vector Review

- **Euclidean norm:** for vector $\vec{x} = (x_1, x_2, \dots, x_n)$,
$$\|\vec{x}\| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2} = \sqrt{\sum_{i=1}^n x_i^2}$$
- **Dot product:** for two $\vec{w}$ and $\vec{x}$,
$$\vec{w} \cdot \vec{x} = w_1 x_1 + w_2 x_2 + \dots + w_n x_n = \sum_{i=1}^n w_i x_i$$
 - Alternative notations: $\vec{w} \cdot \vec{x} = \vec{w}^\top \vec{x} = \langle \vec{w}, \vec{x} \rangle$
- Angle & projection interpretation: $\vec{w} \cdot \vec{x} = \|\vec{w}\| \|\vec{x}\| \cos \theta$
 - $\frac{\vec{w} \cdot \vec{x}}{\|\vec{w}\|}$ is the signed length of the projection of $\vec{x}$ onto $\vec{w}$

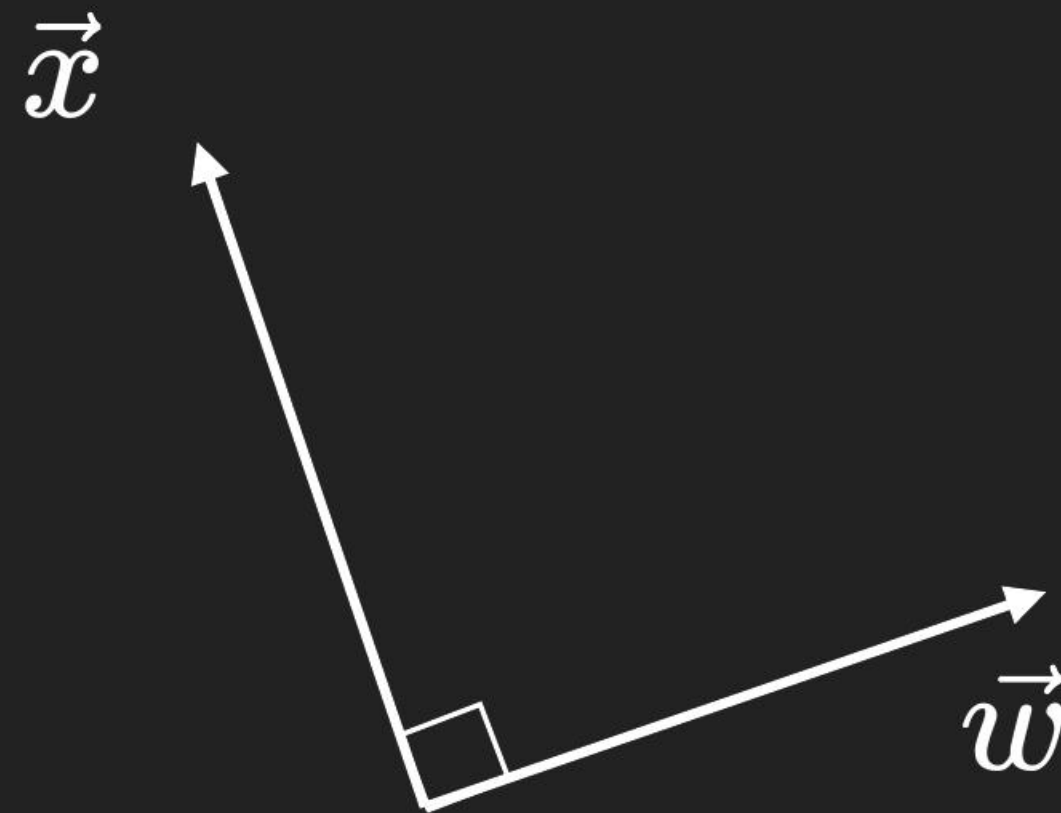


Vector Review

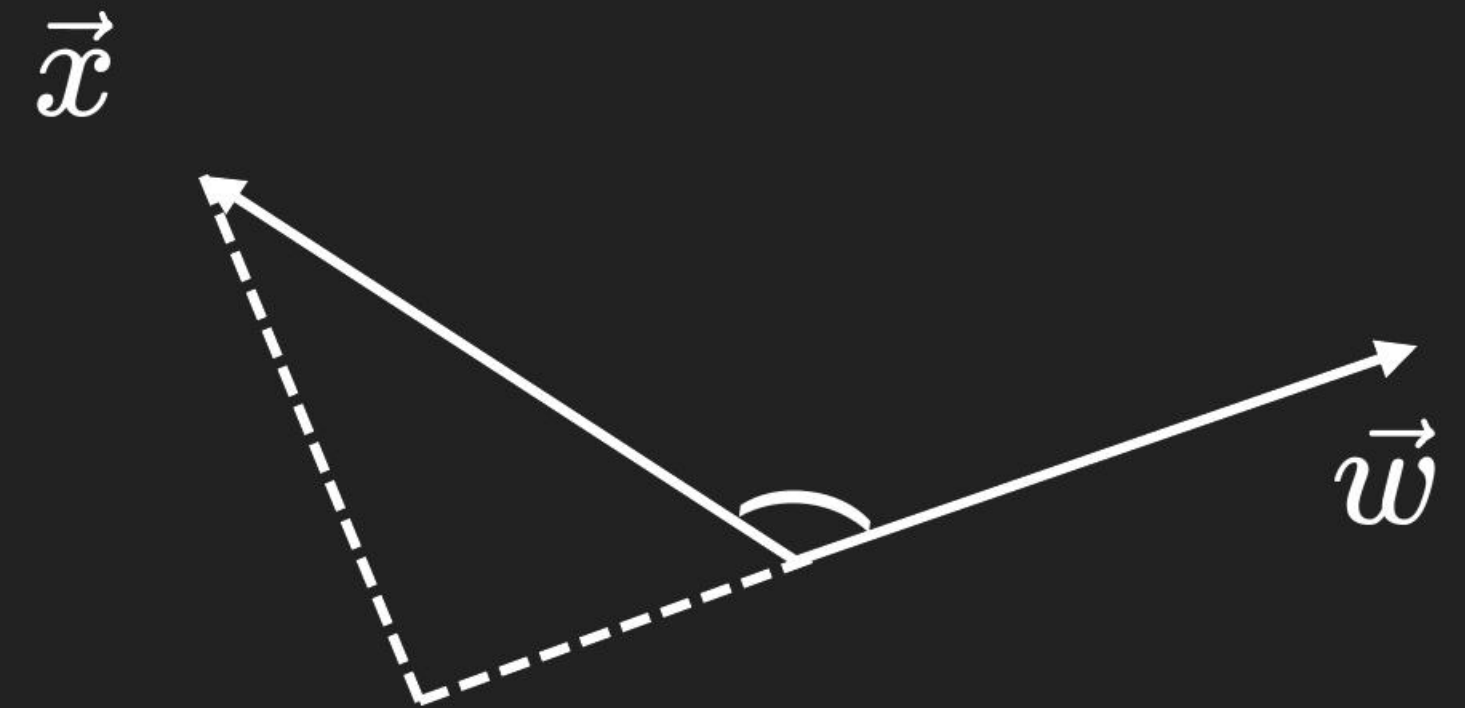
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$$\vec{x} \cdot \vec{w} > 0$$



$$\vec{x} \cdot \vec{w} = 0$$



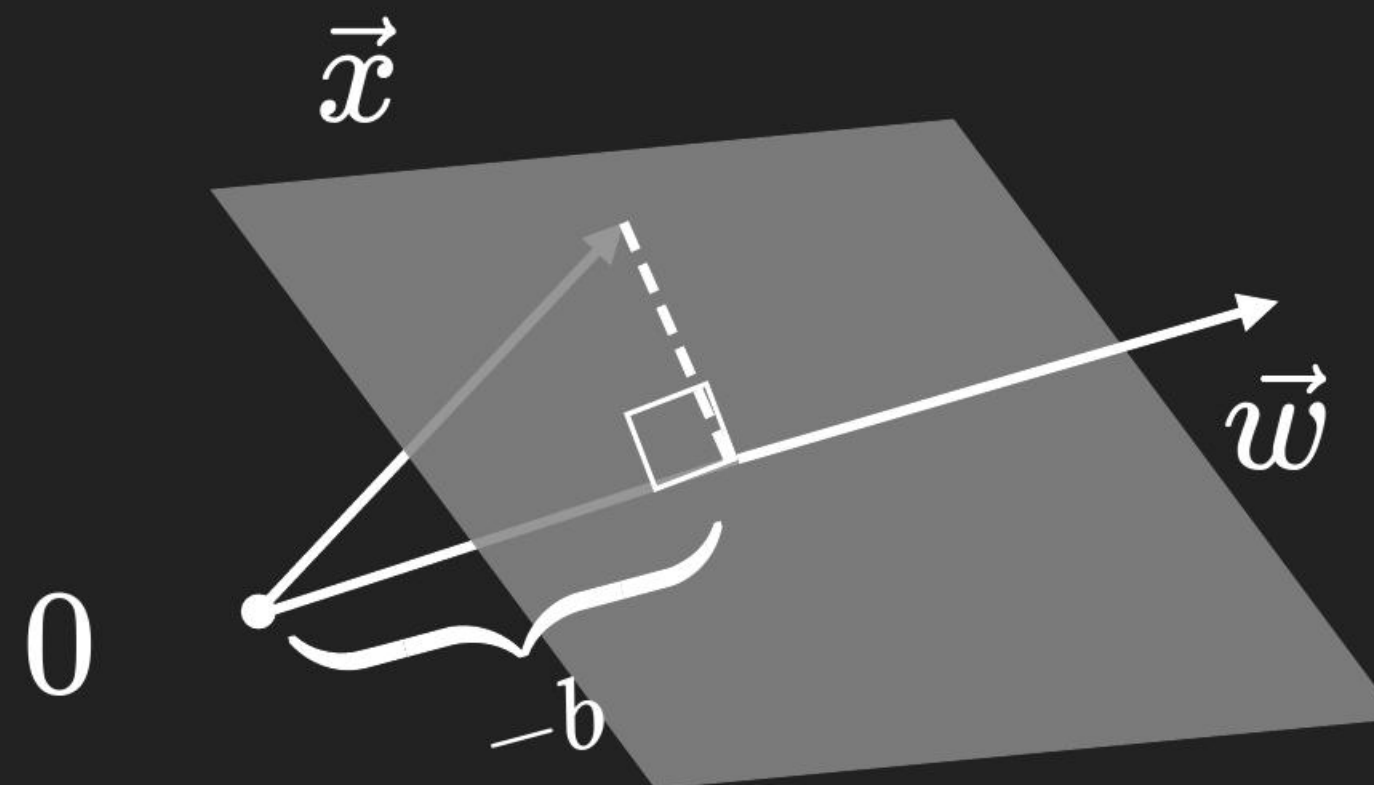
$$\vec{x} \cdot \vec{w} < 0$$

Hyperplanes

- In d dimensional space, a **hyperplane** is defined by a **vector** $\vec{w} \in \mathbb{R}^d$ and a **scalar** $b \in \mathbb{R}$:

$$\{\vec{x} \in \mathbb{R}^d \mid \vec{w} \cdot \vec{x} + b = 0\}$$

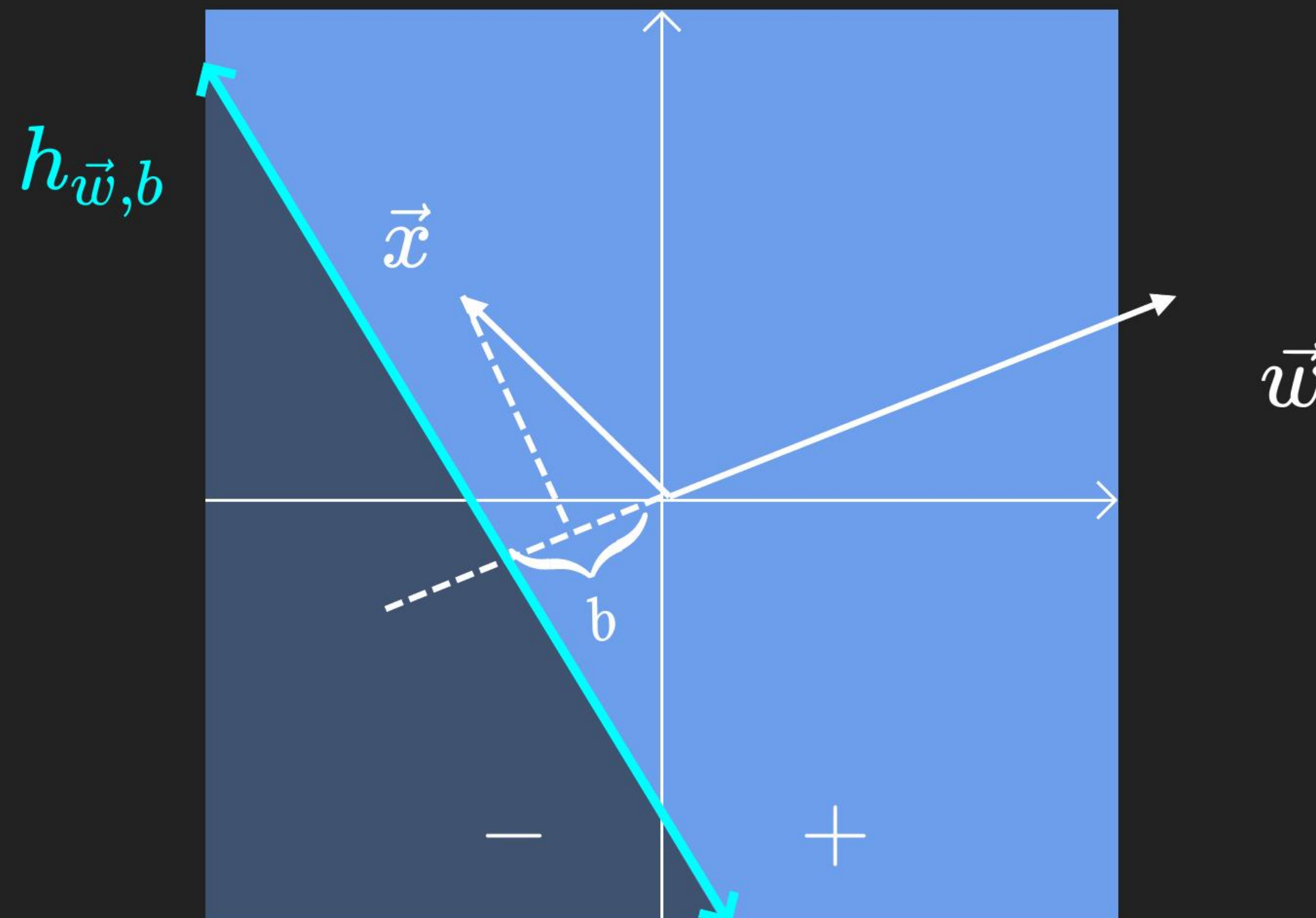
- Geometric interpretation:
 - The hyperplane is orthogonal to $\vec{w}$
 - The distance to the origin (along $\vec{w}$) is $-b/\|\vec{w}\|$
 - All points on hyperplane have $\vec{w} \cdot \vec{x} = -b$



Linear Classifiers

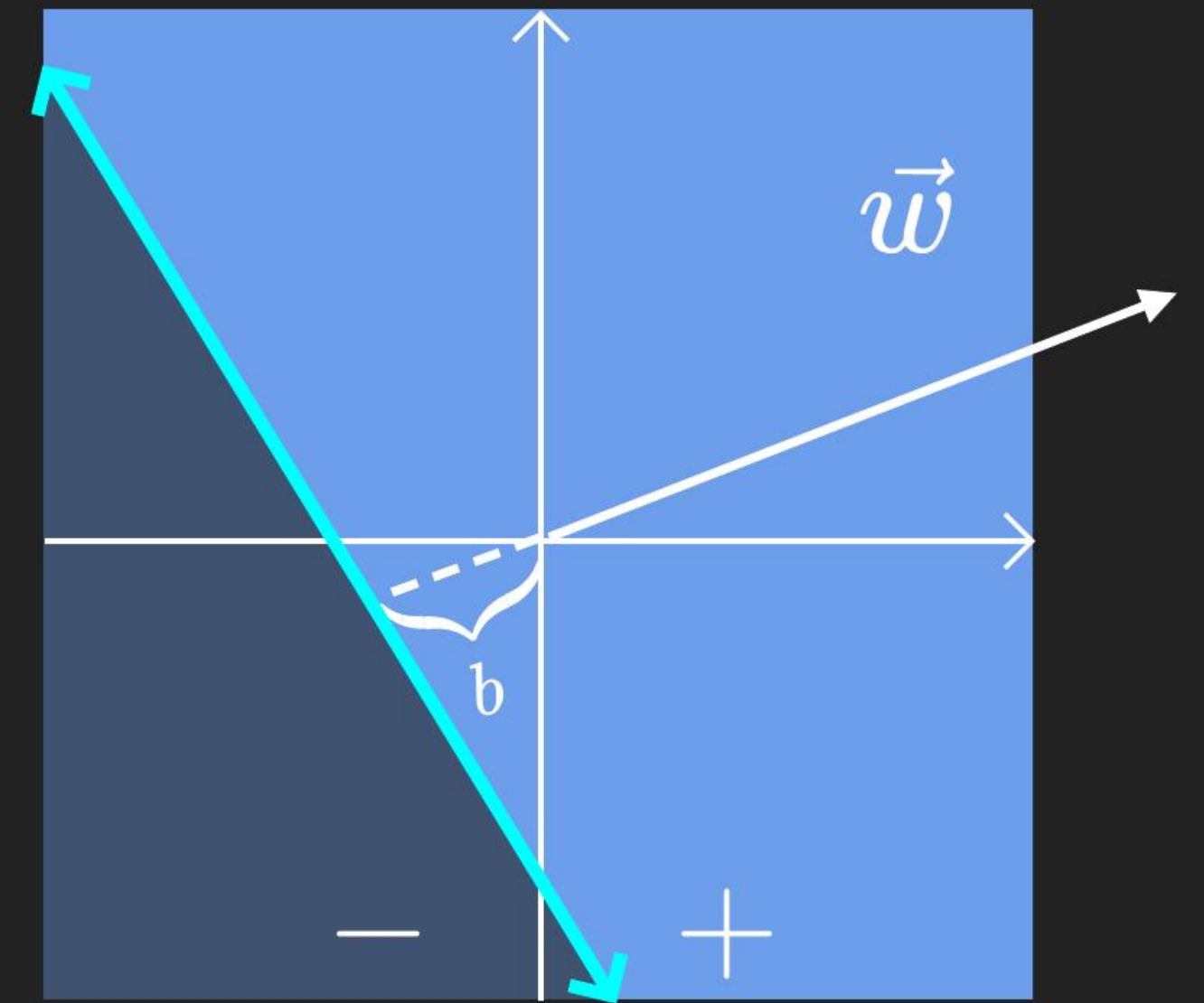
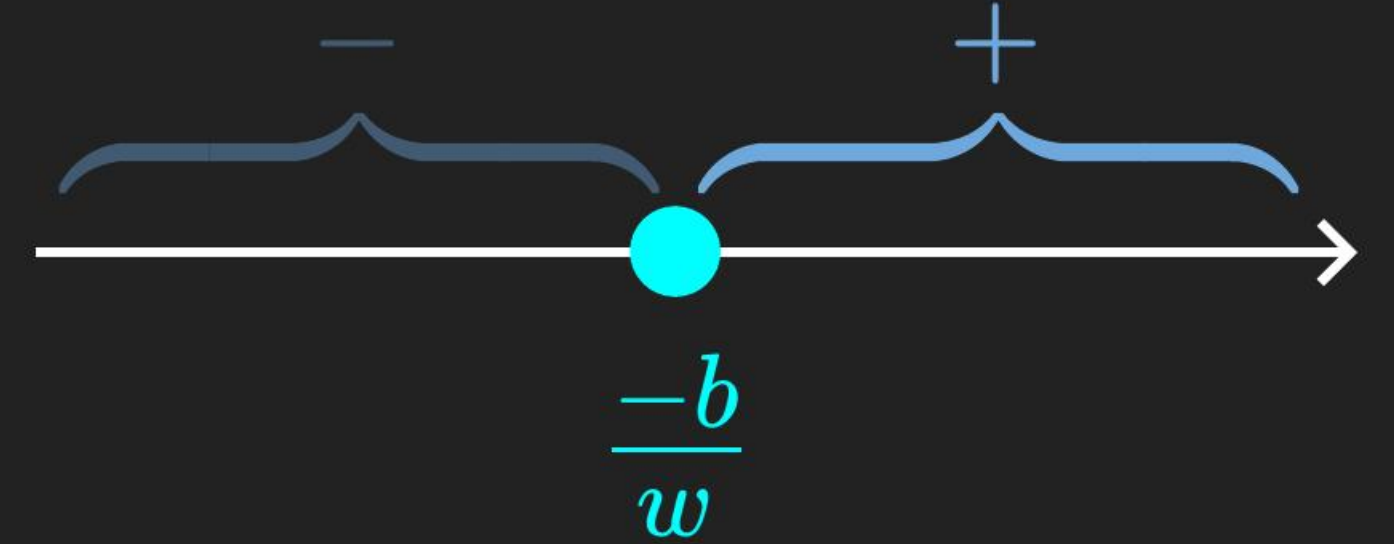
- For a vector $\vec{w} \in \mathbb{R}^d$ and $b \in \mathbb{R}$, the hypothesis $h_{\vec{w},b} : \mathbb{R}^d \rightarrow \{-1, +1\}$ is called a d dimensional linear classifier and defined

$$h_{\vec{w},b}(\vec{x}) = \text{sign}(\vec{w} \cdot \vec{x} + b) = \begin{cases} +1 & \vec{w} \cdot \vec{x} + b > 0 \\ -1 & \vec{w} \cdot \vec{x} + b \leq 0 \end{cases}$$



Linear Classifiers in all Dimensions

- One dimension: $h_{w,b}(x) = \text{sign}(wx + b)$
 - Decision boundary: a point (1d hyperplane)
- Two dimension: $h_{\vec{w},b}(x) = \text{sign}(\vec{w} \cdot \vec{x} + b)$
 - Decision boundary: line (2d hyperplane)
- d dimension: $\text{sign}(\vec{w} \cdot \vec{x} + b)$
 - Decision boundary: hyperplane $\vec{w} \cdot \vec{x} + b = 0$



$$w_1x_1 + w_2x_2 + b = 0$$

Homogenous Linear Classifiers

- A classifier is **homogenous** if $b = 0$ (otherwise non-homogenous)
- **Fact:** any d dimensional learning problem for linear classifiers has a **homogenous** form in $d + 1$ dimensions

Non-homogenous:

- $HS^d = \{h_{\vec{w},b} \mid \vec{w} \in \mathbb{R}^d, b \in \mathbb{R}\}$
 - $\vec{x}$
 - $\vec{w}, b$
 - $\vec{w} \cdot \vec{x} + b$

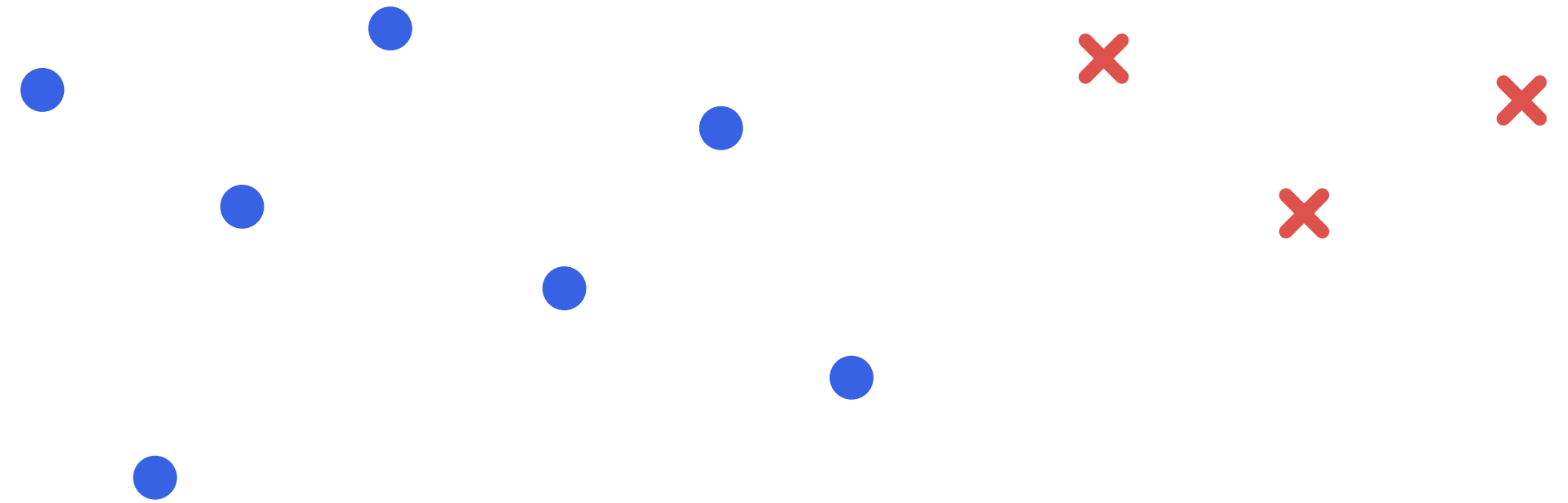
Homogenous:

- $HS_{homog}^{d+1} = \{h_{\vec{w},b} \mid \vec{w} \in \mathbb{R}^{d+1}\}$
 - $\vec{x}' = (\vec{x}, 1)$
 - $\vec{w}' = (\vec{w}, b)$
 - $\vec{w}' \cdot \vec{x}' = \vec{w} \cdot \vec{x} + b$

- Without loss of generality, we will now focus on **homogenous linear classifiers with** $\|\vec{w}_i\| = 1$

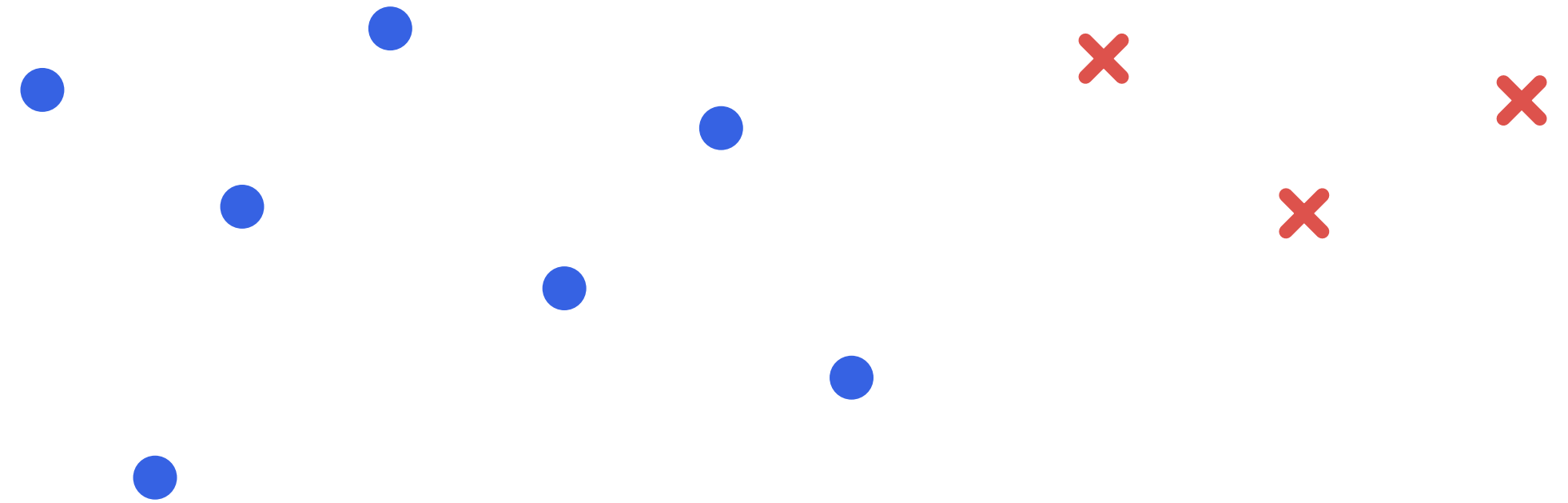
Supervised Learning Setup

- Features $x \in \mathcal{X}$
- Labels $y \in \mathcal{Y}$
- Data $D = \{(x_1, y_1), \dots, (x_n, y_n)\}$



Supervised Learning Setup

- Features $x \in \mathcal{X}$
- Labels $y \in \mathcal{Y}$
- Data $D = \{(x_1, y_1), \dots, (x_n, y_n)\}$
- Model Class $\mathcal{H} = \{h : \mathcal{X} \rightarrow \mathcal{Y}\}$
- Loss Function $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}$
- Learning Algorithm that picks an $h \in \mathcal{H}$ based on D and ℓ



Data

- Data $D = \{(x_1, y_1), \dots, (x_n, y_n)\}$ where $x_i \in \mathbb{R}^d$ are d measurements of data point i . E.g.,
 - Images: raw pixel intensity values flattened into a vector.
 - Text: $x_{i,k} = \begin{cases} 1 & \text{if document } i \text{ contains word } k \\ 0 & \text{otherwise} \end{cases}$

Data

- Data $D = \{(x_1, y_1), \dots, (x_n, y_n)\}$ where $x_i \in \mathbb{R}^d$ are d measurements of data point i . E.g.,
 - Images: raw pixel intensity values flattened into a vector.
 - Text: $x_{i,k} = \begin{cases} 1 & \text{if document } i \text{ contains word } k \\ 0 & \text{otherwise} \end{cases}$
- Labels $y_i \in \mathcal{Y}$ could be, e.g.,
 - Binary classification: $\mathcal{Y} = \{0, 1\}$
 - Multiclass classification: $\mathcal{Y} = \{0, \dots, K\}$
 - Regression: $\mathcal{Y} = \mathbb{R}$

Loss Functions

- Classification Loss: $\ell(y, y') = \begin{cases} 1 & \text{if } y \neq y' \\ 0 & \text{otherwise (} y = y' \text{)} \end{cases}$
- Square Loss: $\ell(y, y') = \|y - y'\|^2$
- Absolute Loss: $\ell(y, y') = |y - y'|$

Loss Functions

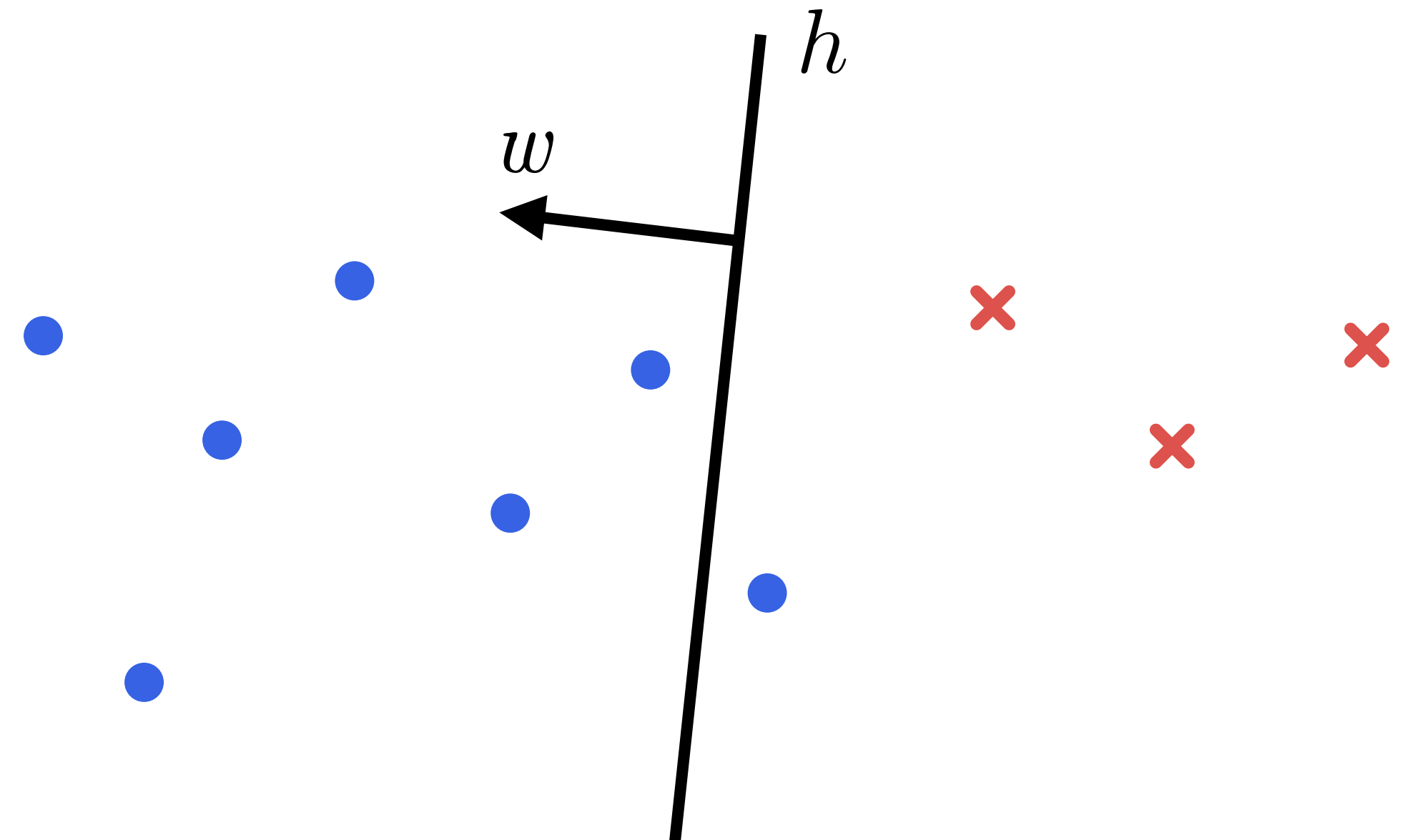
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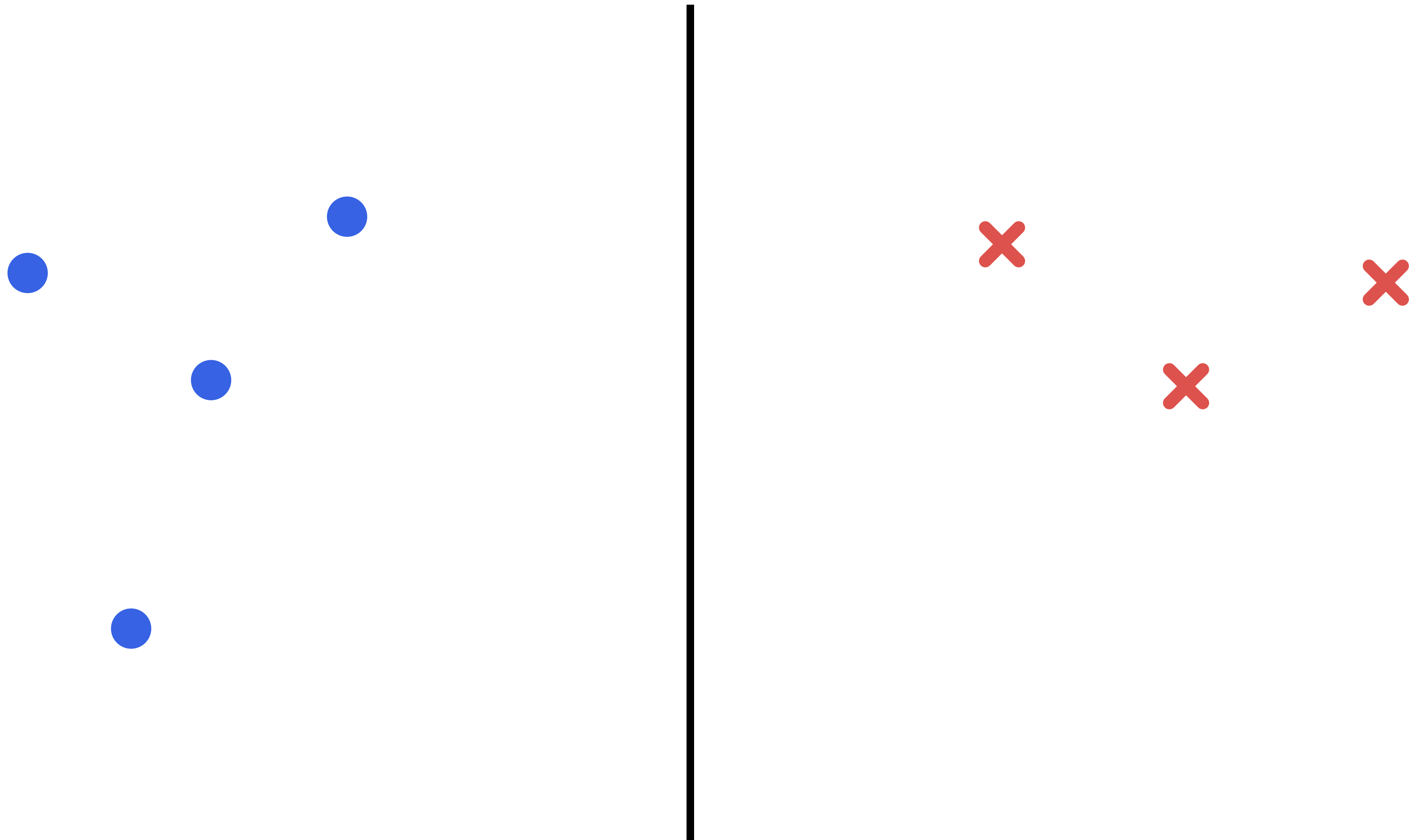
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- Example Classification Loss:

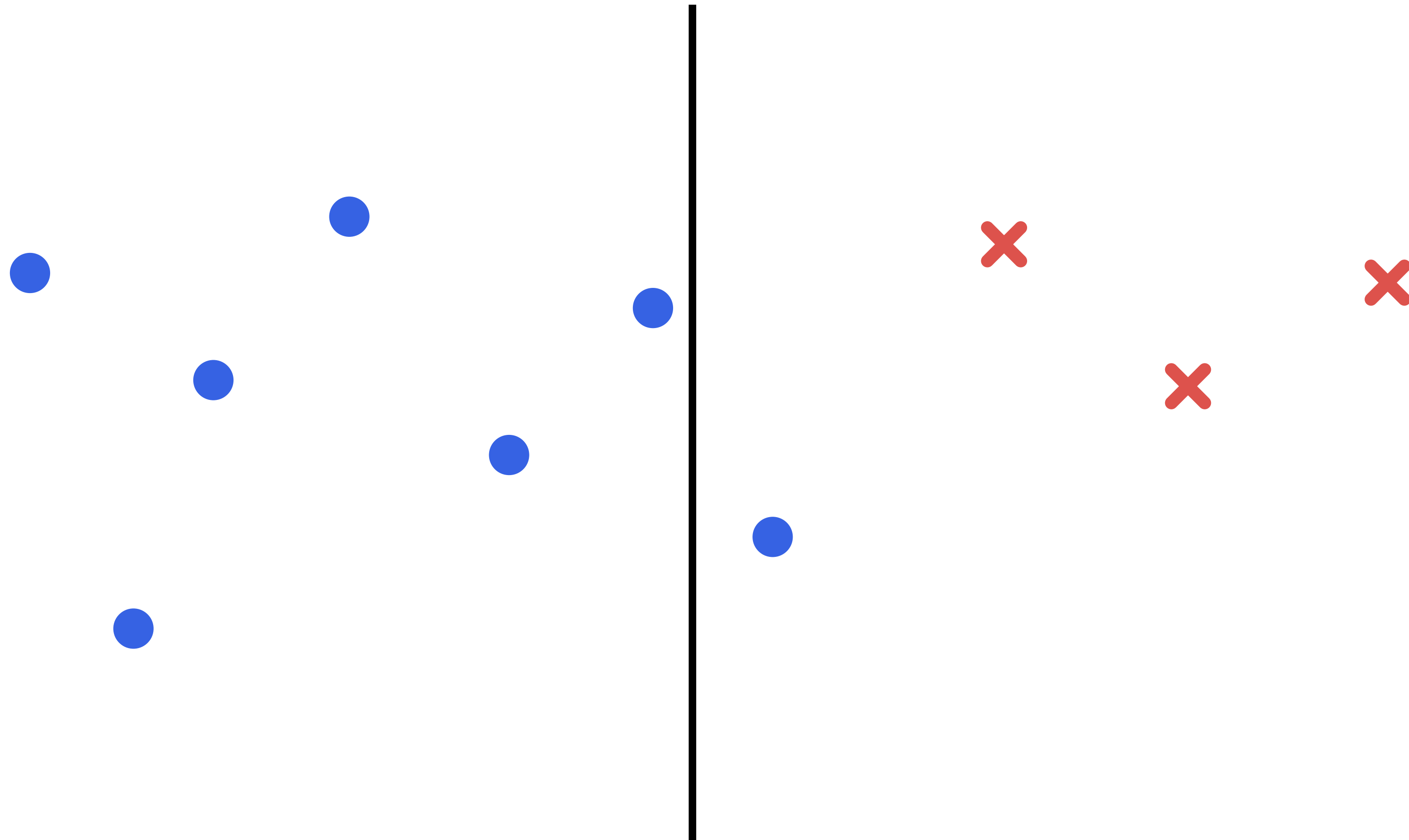
$$\sum_{i=1}^n \ell(y_i, h(x_i))$$



Generalization



Generalization



Can we find a hypothesis that *generalizes* to new data?

Train / Test Split

- Can we *measure* how well a hypothesis generalizes to new data?
- Randomly split the data into two parts: $\mathcal{D} = \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{test}}$
- Choose a hypothesis (somehow) based on $\mathcal{D}_{\text{train}}$
- Evaluate the hypothesis on $\mathcal{D}_{\text{test}}$:

$$\frac{1}{n} \sum_{i=1}^n \ell(y_i, h(x_i)) \quad \text{for } (x_i, y_i) \in \mathcal{D}_{\text{test}}$$